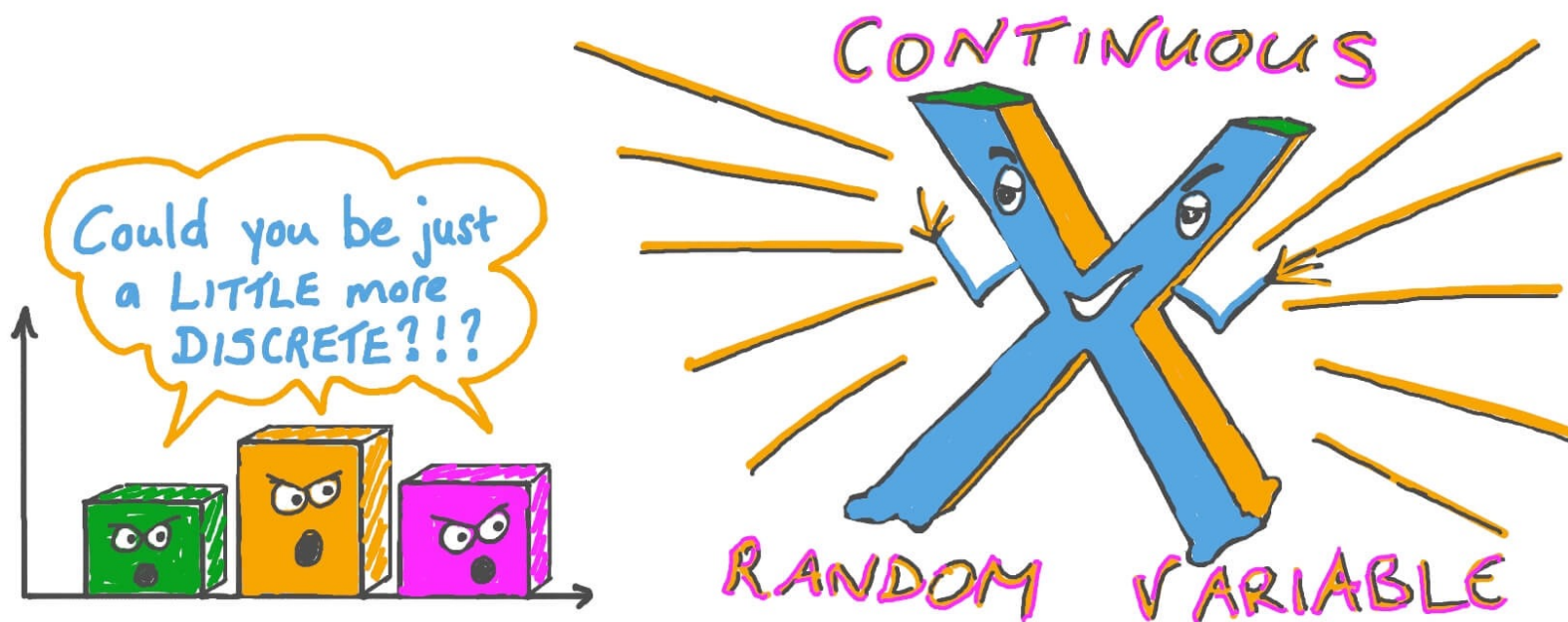


CS 237: PROBABILITY IN COMPUTING

INSTRUCTOR: TIAGO JANUARIO, PH.D.

BOSTON
UNIVERSITY

CONTINUOUS RANDOM VARIABLES



LECTURE 6

Last time

- Quiz 1
- More sample spaces
- Introduction to random variables

Today

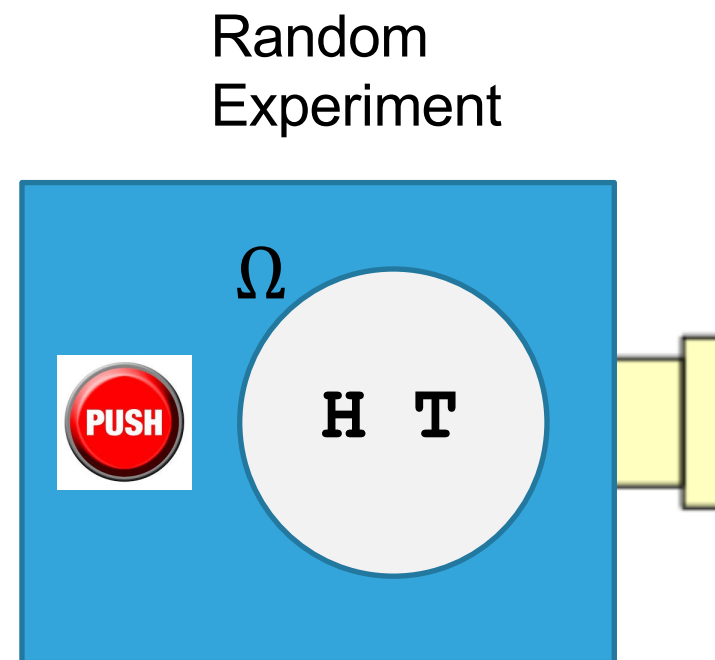
- Random Variables
- Probability Distributions

BEYOND EVENTS

- ▶ In many settings, we would like to know **more than the probabilities** of events
 - ▶ How many days will a system run before it fails?
 - ▶ How many people support candidate A?
 - ▶ How much money will I make if I invest in this stock?
 - ▶ How many heads in n coin tosses?
 - ▶ How many die rolls until we see a 6?
 - ▶ How much time will it take me to mine a Bitcoin on my laptop?

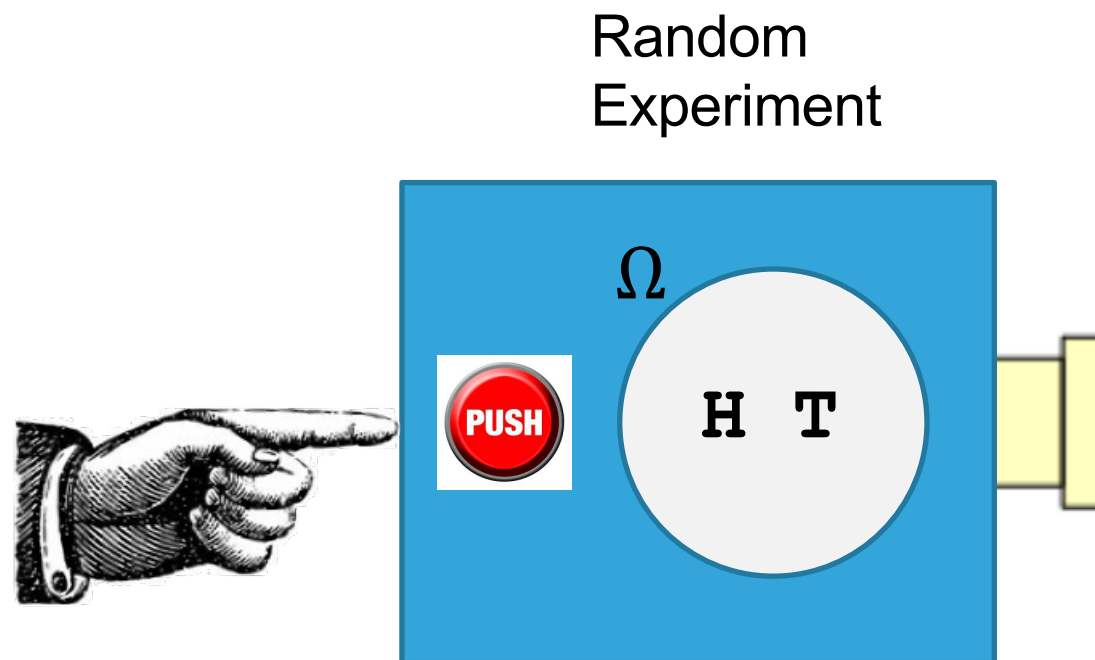
RANDOM EXPERIMENT

- ▶ A **Random Experiment** is a repeatable procedure that produces uncertain outcomes from a well-defined sample space.
- ▶ **Example: Flip a coin!**



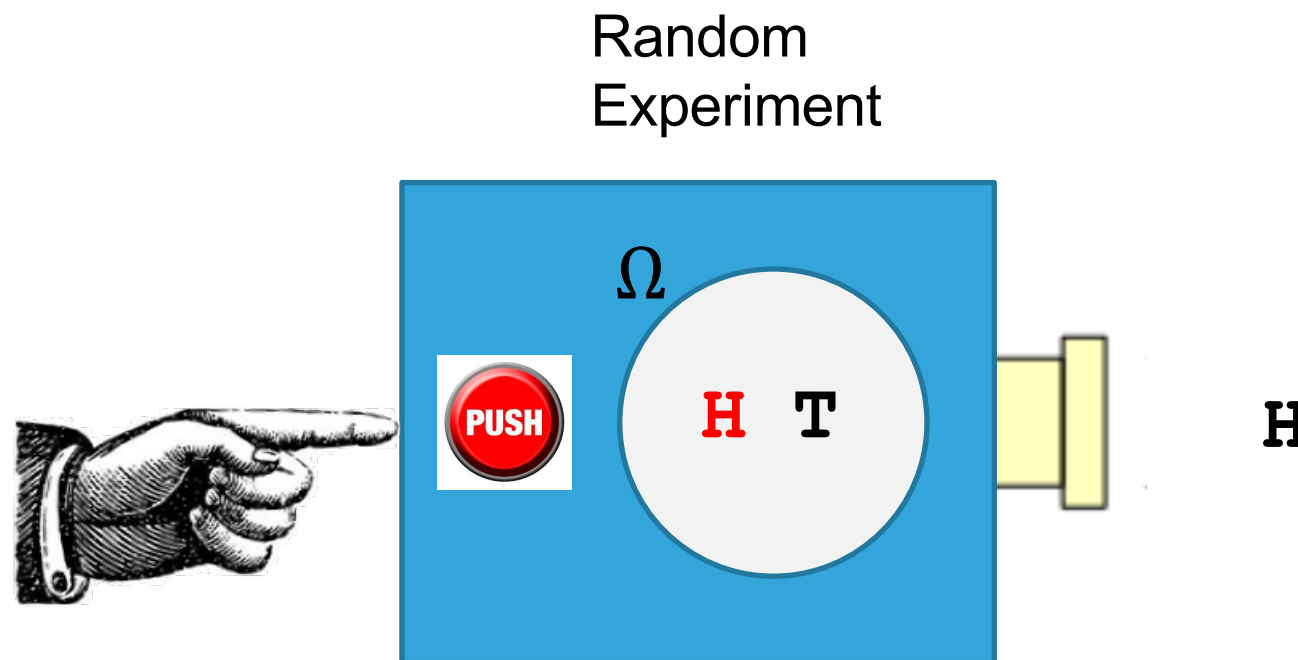
RANDOM EXPERIMENT

- ▶ A **Random Experiment** is a repeatable procedure that produces uncertain outcomes from a well-defined sample space.
- ▶ **Example: Flip a coin!**



RANDOM EXPERIMENT

- ▶ A **Random Experiment** is a repeatable procedure that produces uncertain outcomes from a well-defined sample space.
- ▶ **Example: Flip a coin!**



RANDOM VARIABLES: DEFINITION

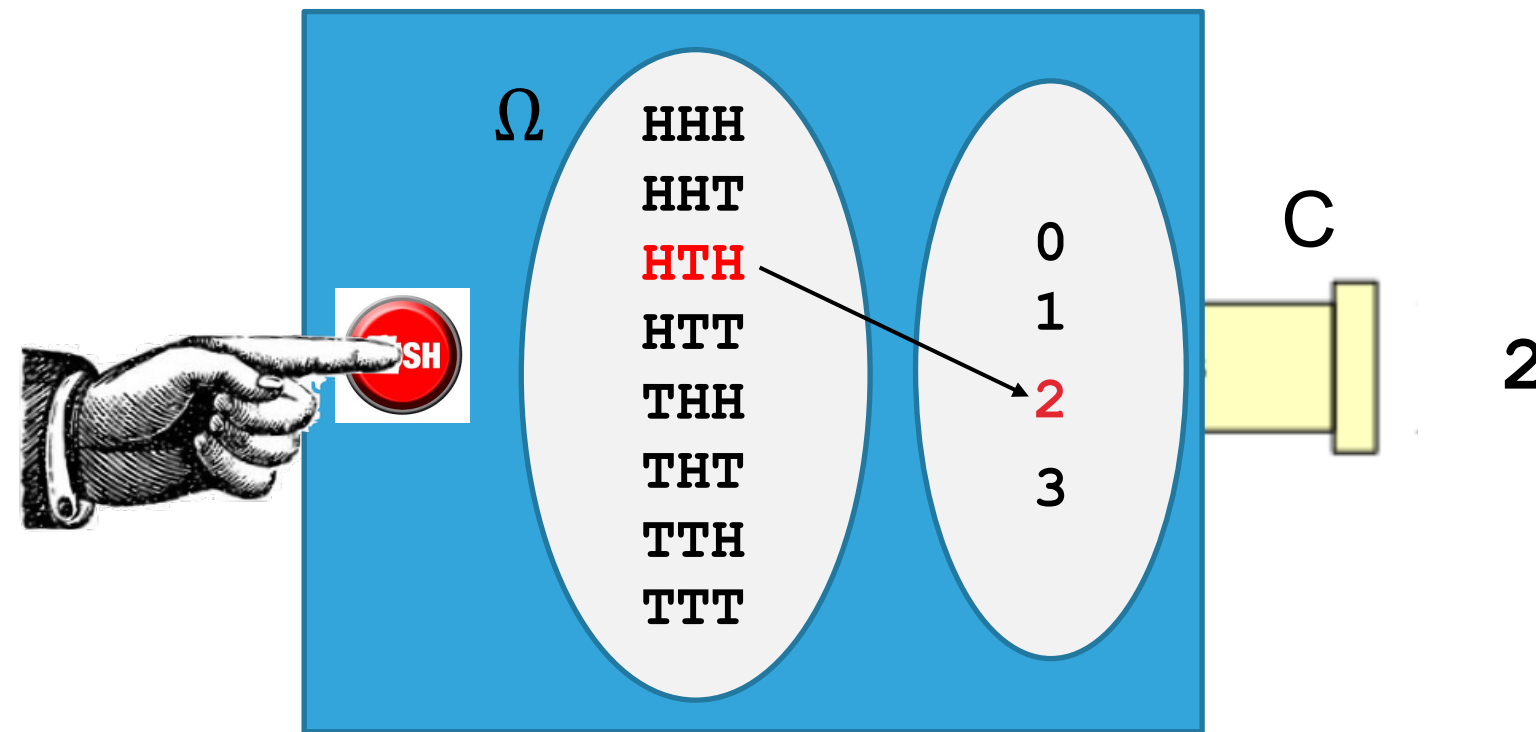
Definition: A random variable R is a function whose domain is the sample space and co-domain is a subset of $\mathbb{R}$:

$$R: \Omega \rightarrow \mathbb{R}$$

- ▶ R maps each outcome $\omega \in \Omega$ to a value $R(\omega) \in \mathbb{R}$
- ▶ Now when an outcome is requested, the sample point is translated into a real number
- ▶ The name “random variable” is a **misnomer**, random variables are actually functions

RANDOM VARIABLES

- ▶ A random variable will not in general simply rename outcomes with real numbers, it will **compute some useful information about the outcome**.
- ▶ $C = \text{"Flip 3 coins and return the number of heads showing"}$



RANDOM VARIABLES ~~functions~~

- ▶ **Experiment:** toss a fair coin 3 times
- ▶ Let C be the number of heads that appear
- ▶ How should we think about C?

RANDOM VARIABLES

- ▶ **Experiment:** toss a fair coin 3 times
- ▶ Let C be the number of heads that appear
- ▶ How should we think about C ?
- ▶ C maps each outcome to a value

$$C(HHH) = 3 \quad C(HHT) = C(HTH) = C(THH) = 2$$

$$C(TTT) = 0 \quad C(TTH) = C(THT) = C(HTT) = 1$$

RANDOM VARIABLES

- ▶ **Experiment:** toss a fair coin 3 times

$$P_r(C = 0) = 1/8$$

$$P_r(C = 1) = 3/8$$

$$P_r(C = 2) = P_r(\{HHT, HTH, THH\}) = 3/8$$

- ▶ Let C be the number of heads that appear $P_r(C = 3) = 1/8$

- ▶ How should we think about C ?

- ▶ C maps each outcome to a value

$$C(HHH) = 3 \quad C(HHT) = C(HTH) = C(THH) = 2$$

$$C(TTT) = 0 \quad C(TTH) = C(THT) = C(HTT) = 1$$

- ▶ C maps each outcome to a value, i.e., C is a **function** from the sample space to values

$$C: \Omega \rightarrow \{0, 1, 2, 3\}$$

RANGE OF RANDOM VARIABLES

- ▶ Since a random variable is a **function**, we can talk about its **range**.
- ▶ The range of a random variable X , shown by $\text{Range}(X)$ or R_X , is the set of possible values for X .
- ▶ I toss a coin five times. Suppose that I'm interested in the **number of heads**. Let X be a **random variable** whose value is the number of heads.

$$\text{Range}(X)=R_X=\{0,1,2,3,4,5\}$$

ASK THE AUDIENCE

Consider the following experiment:

"I toss a coin until the first heads appears. Let Y be the total number of coin tosses."

- ▶ What is the range of the random variable Y ?:

A. $R_Y = \{H, T\}$

B. $R_Y = \{1, 2, 3, 4, 5, \dots\}$

C. $R_Y = \{\sum \frac{1}{2^n} : n \in N\}$

D. $R_Y = \{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \frac{1}{6}, \dots\}$

E. $R_Y = \{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \frac{1}{64}, \dots\}$

Random variables and Events

RANDOM VARIABLES: NOTATION

$$C \neq 0 = \{TTTT\}$$

$$C \neq 1 = \{HTT, THT, TTH\}$$

- ▶ A very typical notion for probabilities involving a random variable X is:

$\Pr(\text{ < some Boolean expression involving } X \text{ > })$

$$\Pr(C \neq 2) = \Pr(C=0) + \Pr(C=1) + \Pr(C=3)$$

$$= 1 - \Pr(C=2)$$

$$\Pr(X = k) = \Pr_X(k)$$

$$\Pr(X \neq k) = 1 - \Pr_X(k) = 1 - \Pr(X=k)$$

$$\Pr(X \leq k) = \sum_{j \leq a \leq k} \Pr_X(a) = \sum_{a \leq k} \Pr(X=a)$$

$$\Pr(C \leq 1) = \Pr(C=0) + \Pr(C=1)$$

- ▶ Here, $\Pr_X(k)$ means the probability that X returns k

RANDOM VARIABLES AND EVENTS

- ▶ The set of outcomes that map to specific values of an r.v. also defines an event.
- ▶ C = number of heads in three tosses

$$\underbrace{TTT}_{C=0}, \underbrace{HTT, THT, TTH}_{C=1}, \underbrace{HHT, HTH, THH}_{C=2}, \underbrace{HHH}_{C=3}$$

$$Pr(C = 2) = Pr(\{HHT, HTH, THH\}) = 3/8$$

$$Pr(C \leq 1) = Pr(\{TTT, HTT, THT, TTH\}) = 1/2$$

$$Pr(C \leq 1) = Pr(C = 0) + Pr(C = 1)$$

RANDOM VARIABLES AND EVENTS

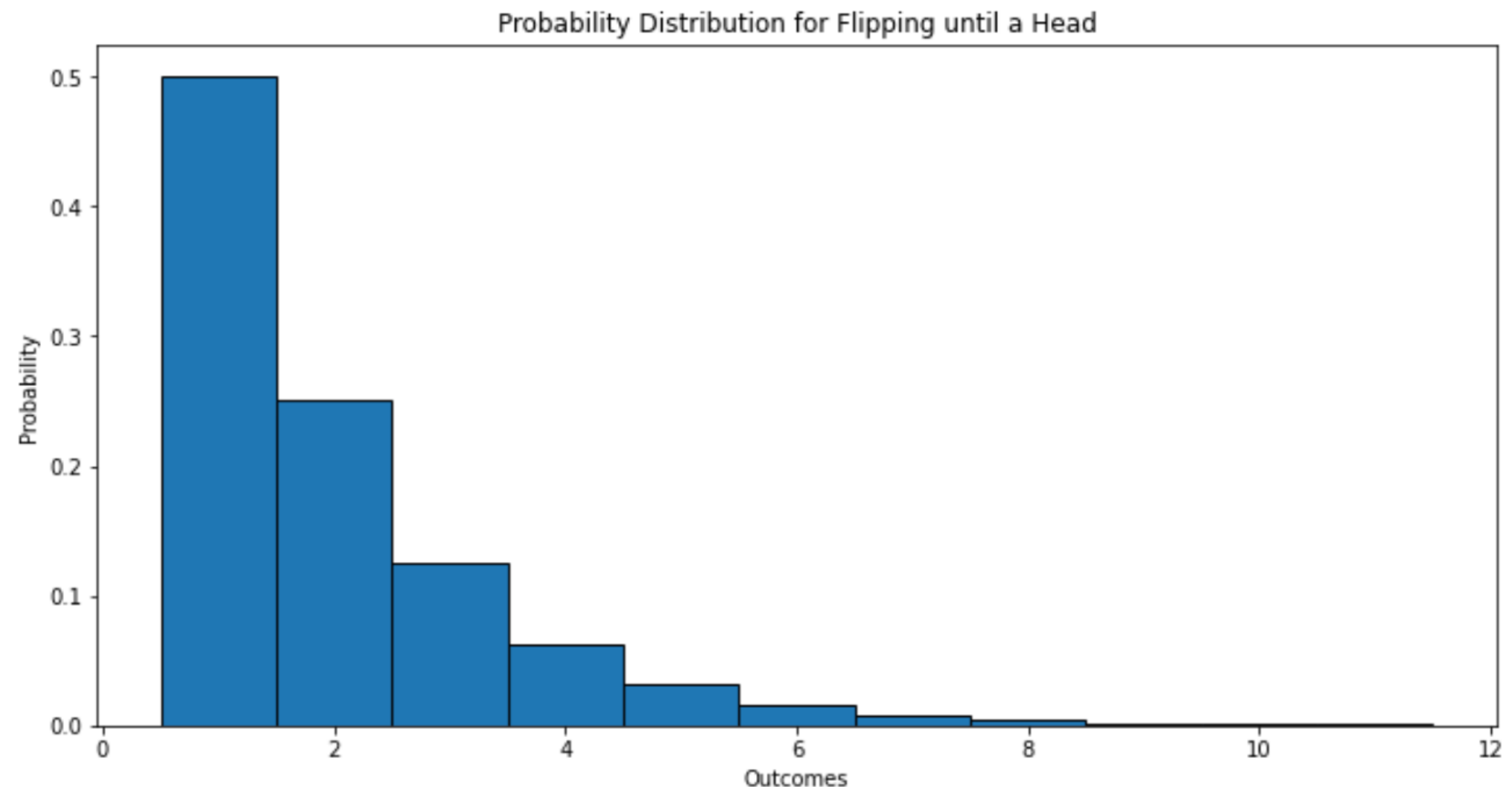
- ▶ Outcomes w/ specific values of an r.v. defines an event
- ▶ C = number of heads in three tosses
- ▶ $M = 1$ if the three tosses match and 0 otherwise
- ▶ $TTT, HTT, THT, TTH, HHT, HTH, THH, HHH$
 $\underbrace{\hspace{1.5cm}}_{C=0} \quad \underbrace{\hspace{2.5cm}}_{C=1} \quad \underbrace{\hspace{2.5cm}}_{C=2} \quad \underbrace{\hspace{1.5cm}}_{C=3}$
- ▶ $HHH, TTT, HTT, THT, TTH, HHT, HTH, THH$
 $\underbrace{\hspace{1.5cm}}_{M=1} \quad \underbrace{\hspace{4.5cm}}_{M=0}$
- ▶ $Pr(\text{both } C \text{ is odd and } M \text{ is odd}) = Pr(\{HHH\}) = \frac{1}{8}$

ASK THE AUDIENCE

For the random variable $X = \text{"Count the number of flips until the coin shows heads"}$

What is $\Pr(X > k)$ for $k > 0$?

- A. $\frac{1}{k}$
- B. $\frac{1}{2^k}$
- C. $\frac{1}{2^{k+1}}$
- D. $\frac{1}{2^{k-1}}$



SOLUTION: Let's solve an alternative problem:

$$Pr(X > k) = 1 - \underline{Pr(X \leq k)} = 1 - Pr(\{X=1\} \cup \{X=2\} \cup \dots \cup \{X=k\})$$

$$\begin{aligned} Pr(X \leq k) &= Pr(X=1) + Pr(X=2) + Pr(X=3) + \dots + Pr(X=k) \\ &= Pr('H') + Pr('TH') + Pr('TTH') + \dots + Pr(\underbrace{'TT \dots TH'}_{(k-1) \text{ times}}) \end{aligned}$$

$$Pr(X \leq k) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^k}$$

SOLUTION

$$\Pr(X > k) = 1 - \Pr(X \leq k) = 1 - \Pr(\{X=1\} \cup \{X=2\} \cup \dots \cup \{X=k\})$$

$$2(\Pr(X \leq k)) = \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^k} \right) \cdot 2$$

$$\Pr(X \leq k) = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \dots + \frac{1}{2^k}$$

$$2\Pr(X \leq k) = 1 + \frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^{k-1}}$$

$$\Pr(X \leq k) = 1 - \frac{1}{2^k}$$

$$\Pr(X > k) = 1 - \left(1 - \frac{1}{2^k} \right) = \frac{1}{2^k}$$

Examples of Random Variables

RANDOM VARIABLES

- ▶ **Discrete random variables**
 - ▶ The possible values (outputs) are listable/countable
 - ▶ Number of H in 3 tosses, sum of two rolls of a die, number of tosses until H, ...
- ▶ **Continuous random variables**
 - ▶ The possible values (outputs) form an interval of $\mathbb{R}$
 - ▶ Spinners, darts, ...

RANDOM VARIABLES: EXAMPLES

- ▶ **Experiment:** toss a fair coin 3 times

$$M = \begin{cases} 1 & \text{if all three tosses match (all H or all T)} \\ 0 & \text{otherwise} \end{cases}$$

Bernoulli
random
variable

$$M: \Omega \rightarrow \{0,1\}$$

$$M(HHH) = M(TTT) = 1 \quad \frac{2}{8}$$

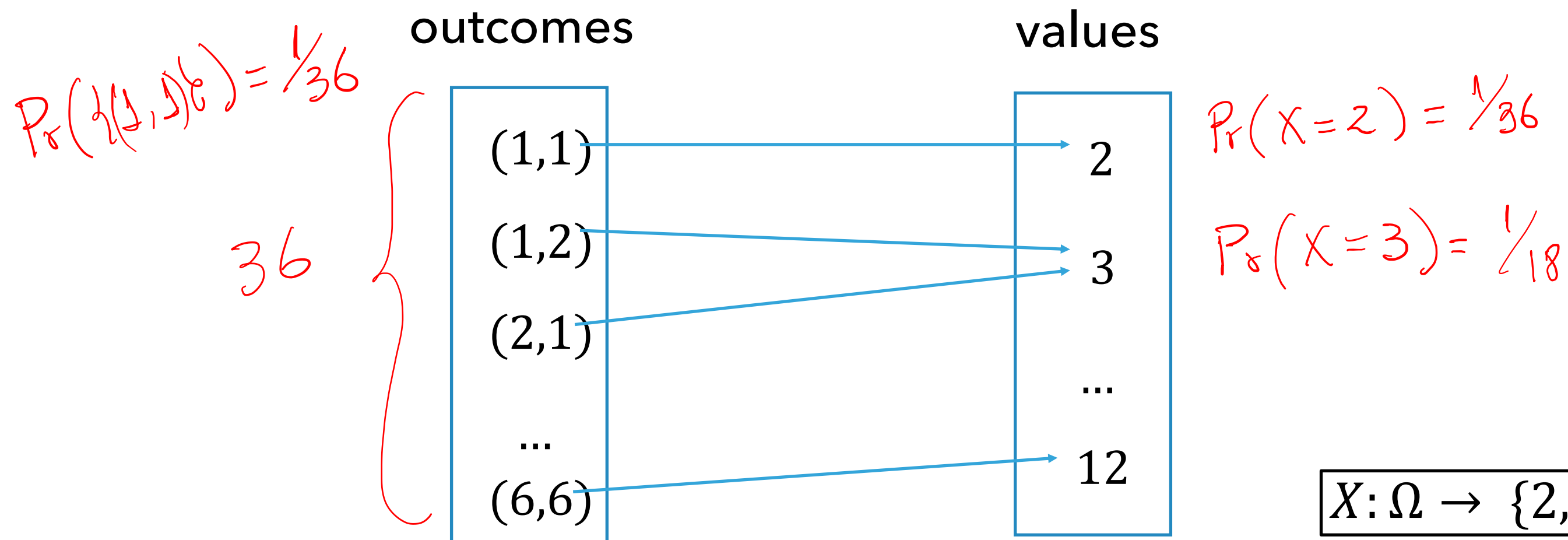
$$M(HHT) = M(HTH) = M(HTT) = 0 \quad \frac{6}{8}$$

$$M(THH) = M(THT) = M(TTH) = 0$$

- ▶ An R.V. whose codomain is $\{0, 1\}$ is an indicator variable.
- ▶ Indicator variables are another way to define *events*.

RANDOM VARIABLES: EXAMPLES

- ▶ **Experiment:** roll a fair die twice $|\Omega| = 36$
- ▶ X = sum of the two rolls



RANDOM VARIABLES: EXAMPLES

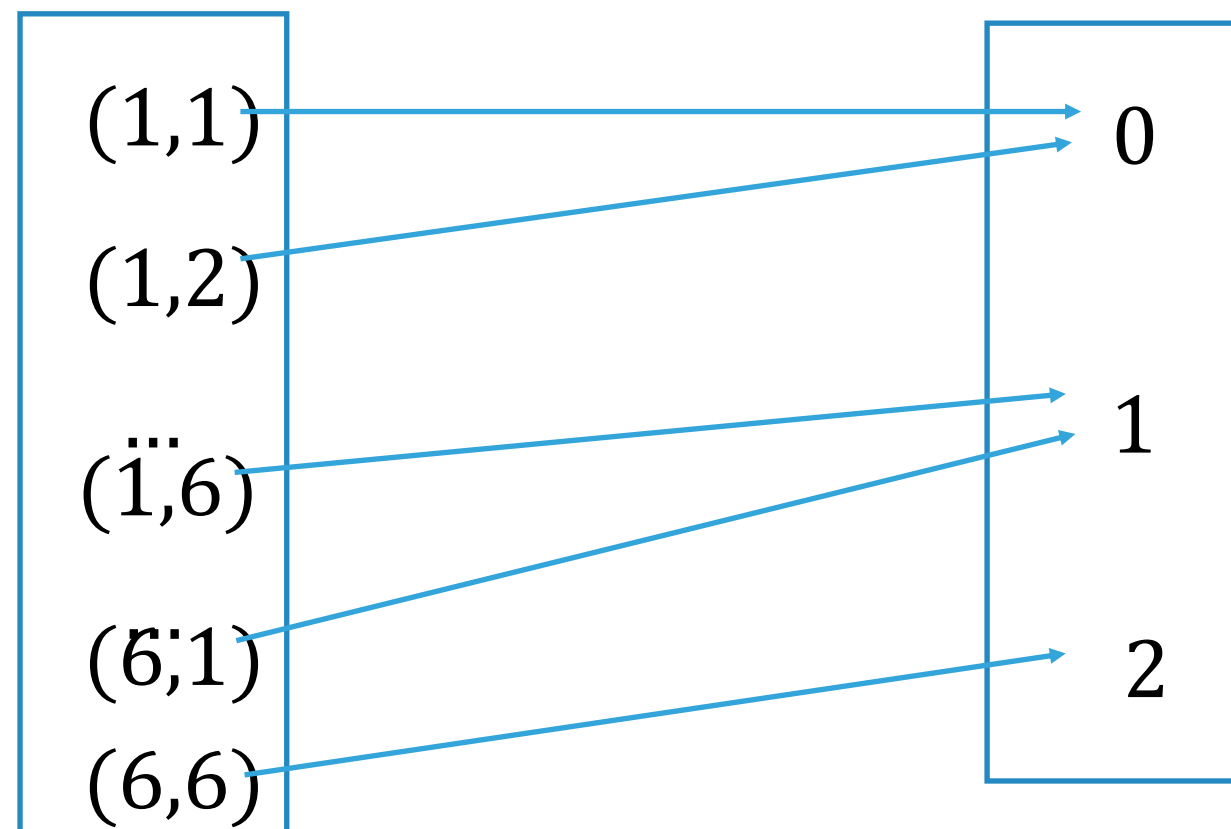
- ▶ **Experiment:** roll a fair die twice
- ▶ Y = number of sixes in the two rolls

	1	2	3	4	5	6
1						
2						
3						
4						
5						
6						

$$\text{Range}(Y) = \{0, 1, 2\}$$

outcomes

values



$$P_0(Y=0) = \frac{25}{36}$$

$$P_0(Y=1) = \frac{10}{36}$$

$$P_0(Y=2) = \frac{1}{36}$$

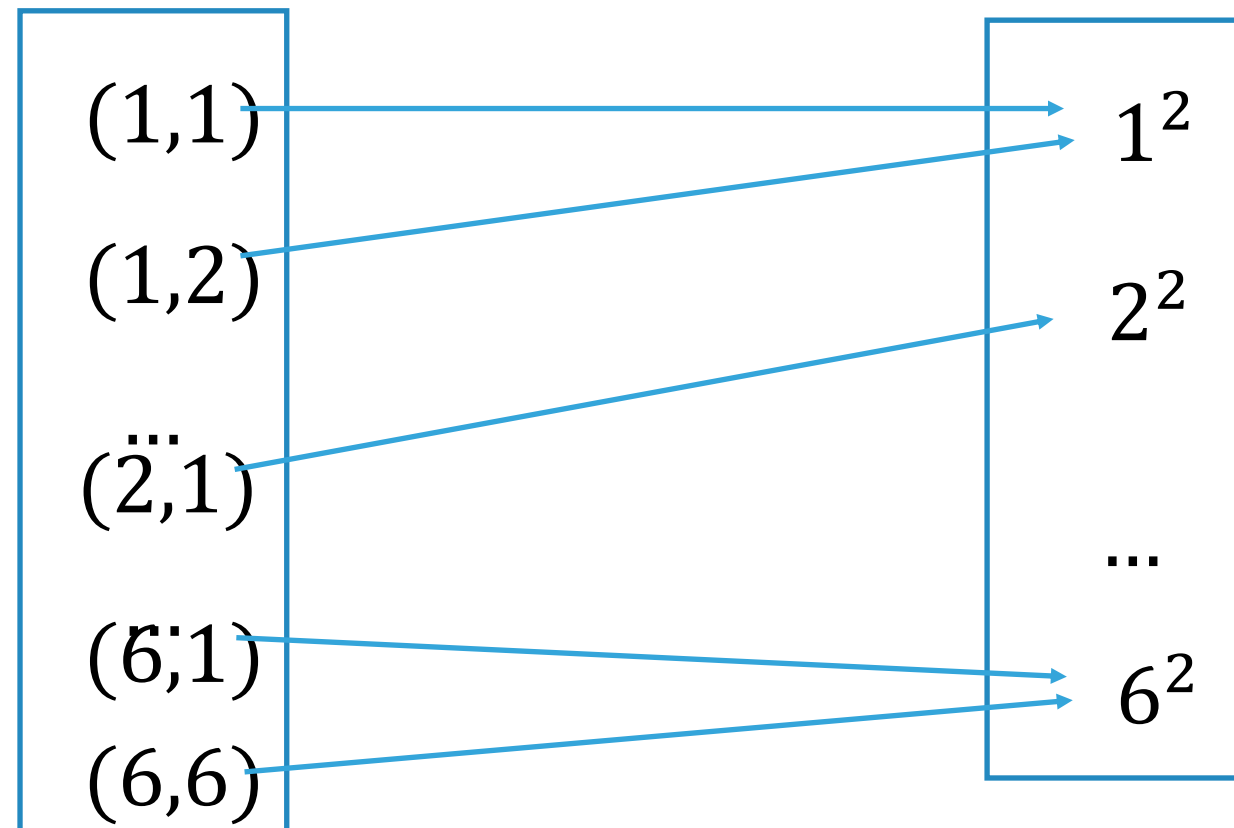
RANDOM VARIABLES: EXAMPLES

- ▶ **Experiment:** roll a fair die twice
- ▶ Z = first roll squared

Range(Z) = {1, 4, 9, 16, 25, 36}

outcomes

values



$\Pr(1^2) =$

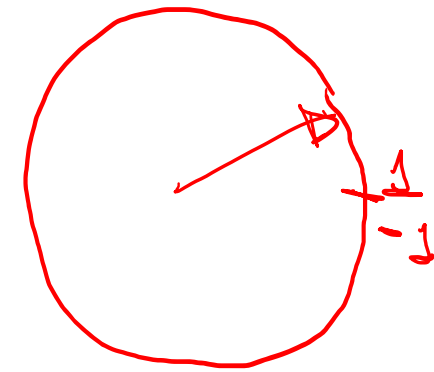
$\vdots$

$\Pr(6^2) =$

RANDOM VARIABLES: EXAMPLES

- **Experiment:** sample a value uniformly from $[-1, 1]$

$$X: [-1, 1] \rightarrow \mathbb{R}, X(r) = r \quad \forall r \in [-1, 1]$$



domain (inputs)

 $[-1, 1]$

values (outputs)

 $[-1, 1]$

RANDOM VARIABLES: EXAMPLES

- ▶ **Experiment:** sample a value uniformly from $[-1, 1]$

$$Y: [-1, 1] \rightarrow \mathbb{R}, Y(r) = r^2 \quad \forall r \in [-1, 1]$$

domain (inputs)

$[-1, 1]$

values (outputs)

$[0, 1]$

RANDOM VARIABLES: EXAMPLES

- **Experiment:** sample a value x uniformly from $[-1, 1]$

$$Z: [-1, 1] \rightarrow \mathbb{R}, Z(r) = \begin{cases} 1 & \text{if } r > 0 \\ 0 & \text{if } r = 0 \\ -1 & \text{if } r < 0 \end{cases}$$

domain (inputs)

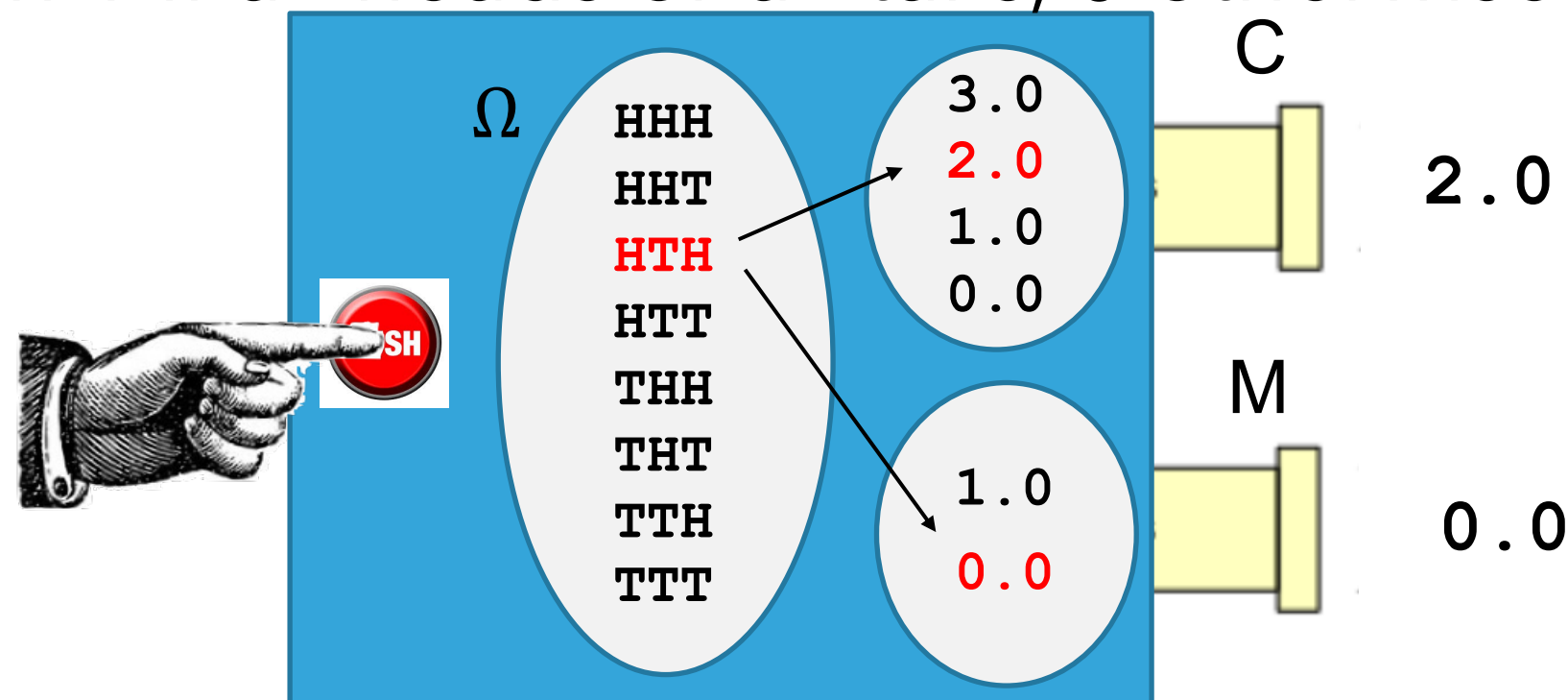
$[-1, 1]$

values (outputs)

$\{1, 0, -1\}$

JOINT RANDOM VARIABLES

- ▶ **Joint Random Variables** are combinations of 2 or more RVs, giving *multiple interpretations* of the same experiment.
- ▶ Example (from LLM):
- ▶ "Flip 3 coins and
(C) return the number of heads showing;
(M) return 1 if all heads or all tails, 0 otherwise"



ASK THE AUDIENCE

Consider the following experiment:

“Flip 3 coins and

(T) return the number of tails showing; 1

(V) return 1 if there are more heads showing than tails, 0 if more tails than heads 1

- Suppose the random experiment chooses **HHT** from Ω then the result is:

A. $T = 1$ $V = 0$

B. $T = 2$ $V = 0$

C. $T = 2$ $V = 1$

D. None of the above

E. $T = 1$ $V = 1$

HE'S ON FIRE

WIKIPEDIA
The Free Encyclopedia

The “hot hand” is the purported phenomenon that a person who experiences a successful outcome with a random event has a greater probability of success in further attempts. The concept is often applied to sports and skill-based tasks in general and originates from basketball, whereas a shooter is allegedly more likely to score if their previous attempts were successful, i.e. while having “hot hands.”

https://en.wikipedia.org/wiki/Hot_hand

HOT HANDS

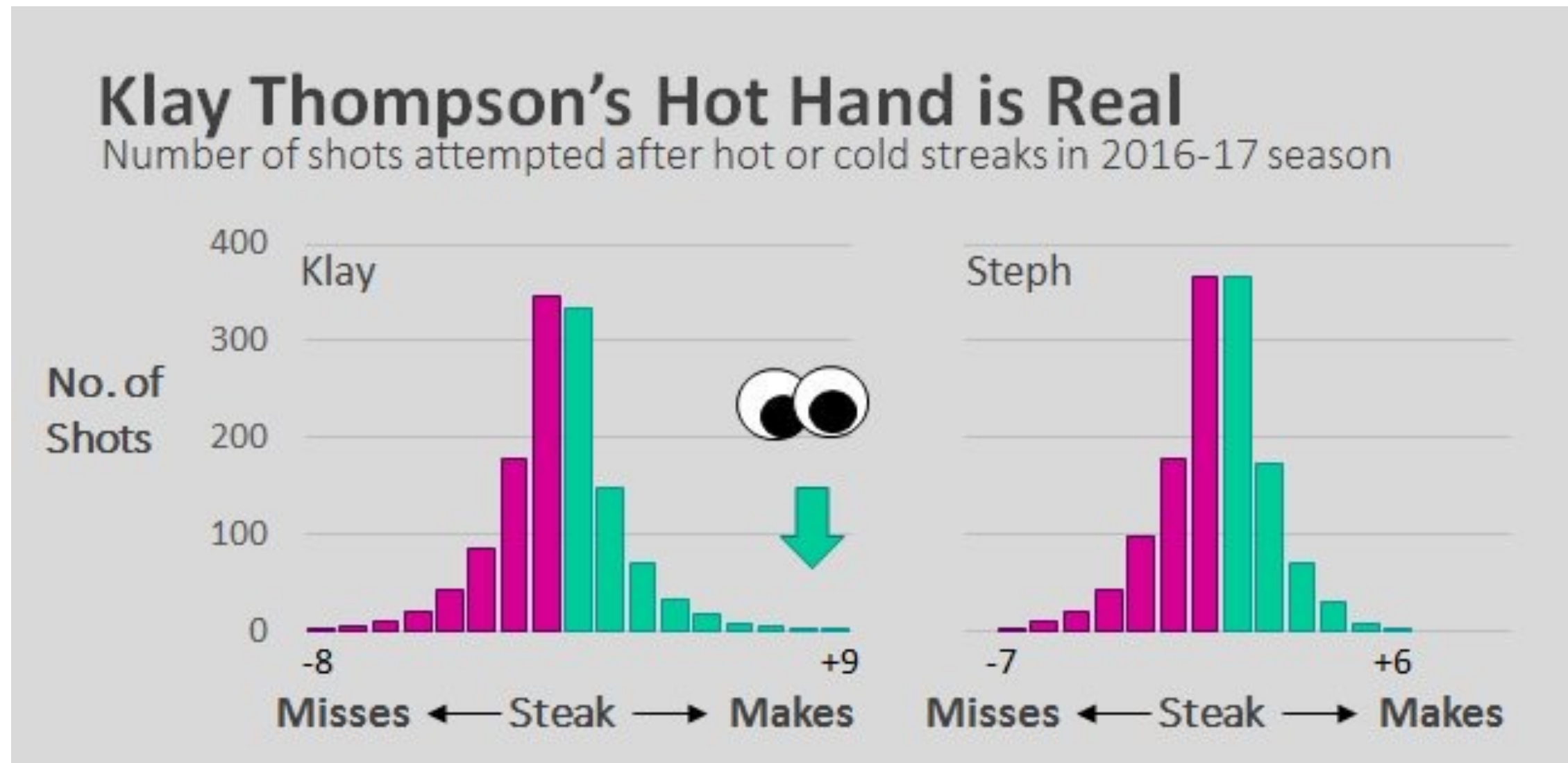
- ▶ **Experiment:** toss a fair coin n times
- ▶ Goal: understand the probability of patterns such as streaks of Heads or Tails
 - ▶ Let us consider the following random variable:
- ▶ HotHand = the length of the longest streak of Heads
- ▶ $\text{HotHand} = \Omega \rightarrow \{0, 1, 2, \dots, n\}$
- ▶ Example ($n=10$): $\text{HotHand}(\text{TTHHHTHTHH}) = 3$

HE'S ON FIRE

- ▶ Klay Thompson's record-setting game
- ▶ Golden State Warriors vs. Sacramento Kings (Jan 23, 2015)
- ▶ Klay Thompson sets several NBA records in 3rd quarter
 - ▶ Most number of pts in a single quarter (37 pts)
 - ▶ Most number of 3 pointers (9)
 - ▶ ...

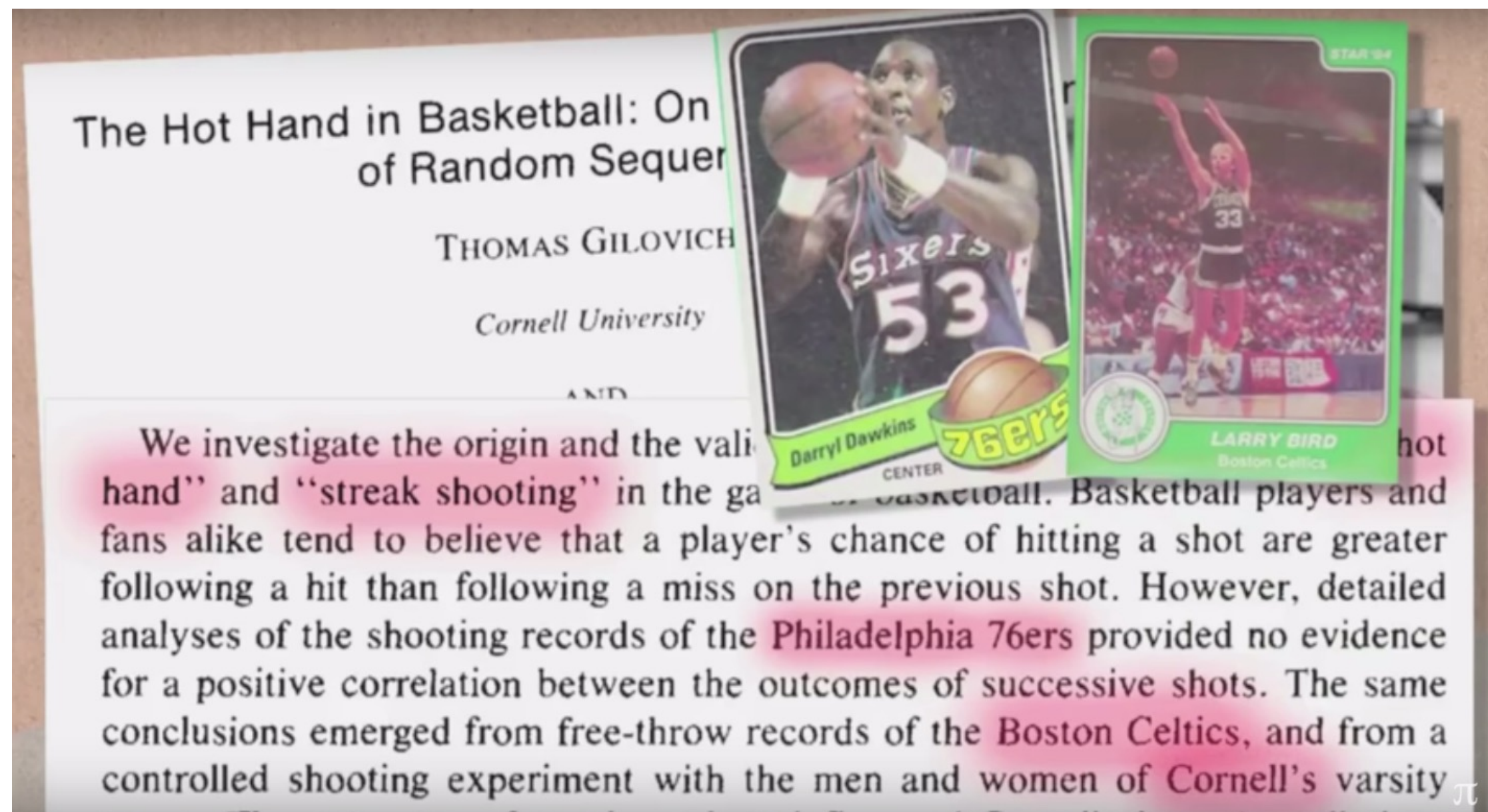


HE'S ON FIRE



From <https://twitter.com/CrumpledJumper/status/873289654505570304>

HOT HANDS



<https://www.youtube.com/watch?v=bPZFQ6i759g>

HOT HANDS

- ▶ Several academic studies have studied the hot hand phenomenon in basketball and other sports
- ▶ A classical study by Gilovich, Vallone, and Tversky (1985) analyzed shooting records of several teams and concluded that there is no evidence for **hot** (or **cold**) hands in basketball
- ▶ There is currently some debate in the academic community about their methodology, corrections have been proposed
- ▶ Basketball fans tend to dismiss these academic findings (but have you seen him score last night? he made 5 shots in a row!)

HOT HANDS

- ▶ Regardless of whether hot/cold hand effects exist in sports, we can study random sequences and see what we find
- ▶ **Experiment:** toss a fair coin n times
- ▶ Goal: understand the probability of patterns such as streaks of Heads (hot hands) or Tails (cold hands)
 - ▶ HotHand = the length of the longest streak of Heads
 - ▶ ColdHand = the length of the longest streak of Tails

ASK THE AUDIENCE

- We toss 10 times. Which event is most likely?

(A) HotHand = 1

HTHTHT

(B) HotHand = 2

HTHTHTHT

(C) HotHand = 3

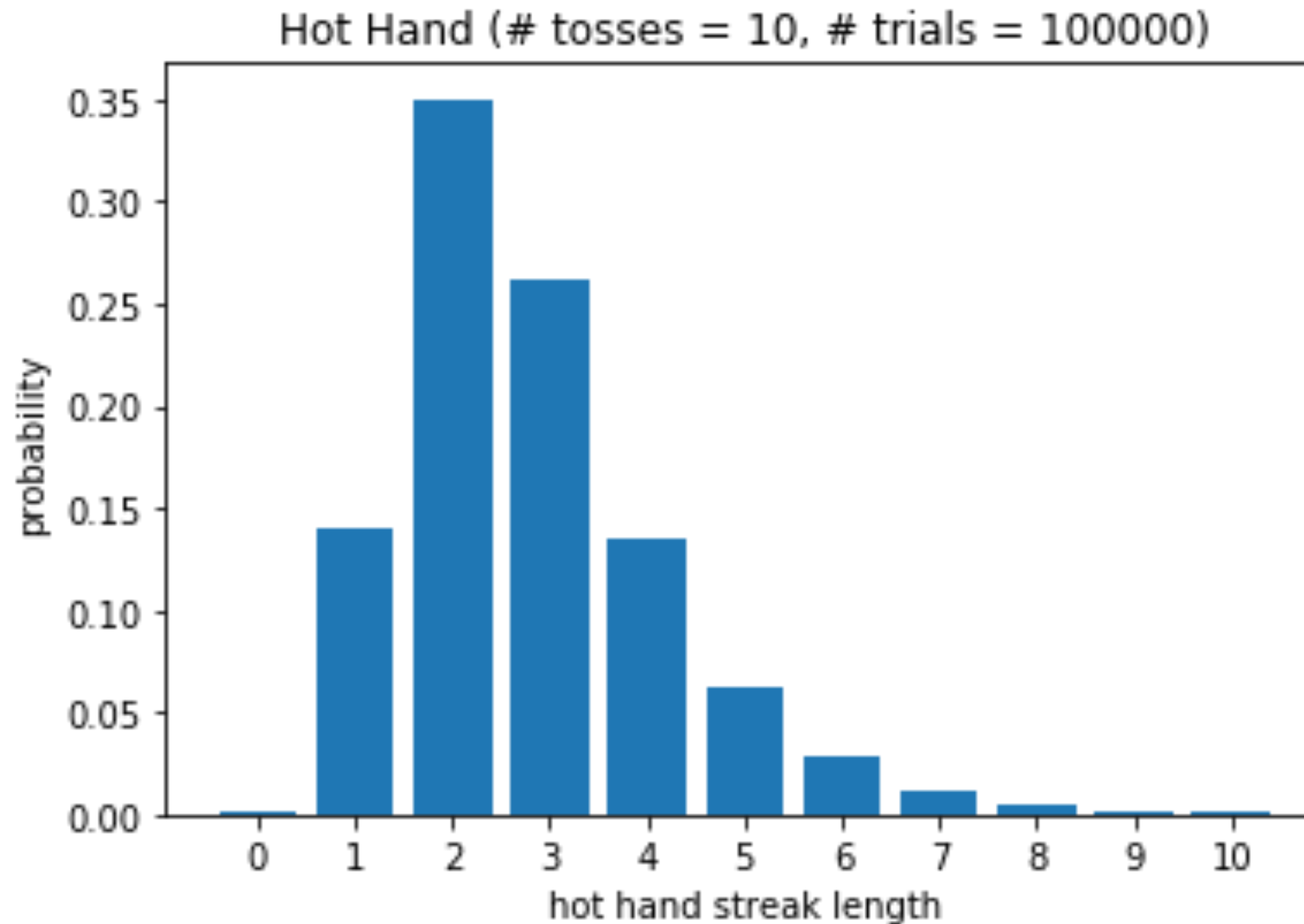
HTHTHTHTHT

(D) HotHand = 4

THTHTHT

1

HOT HAND STREAKS



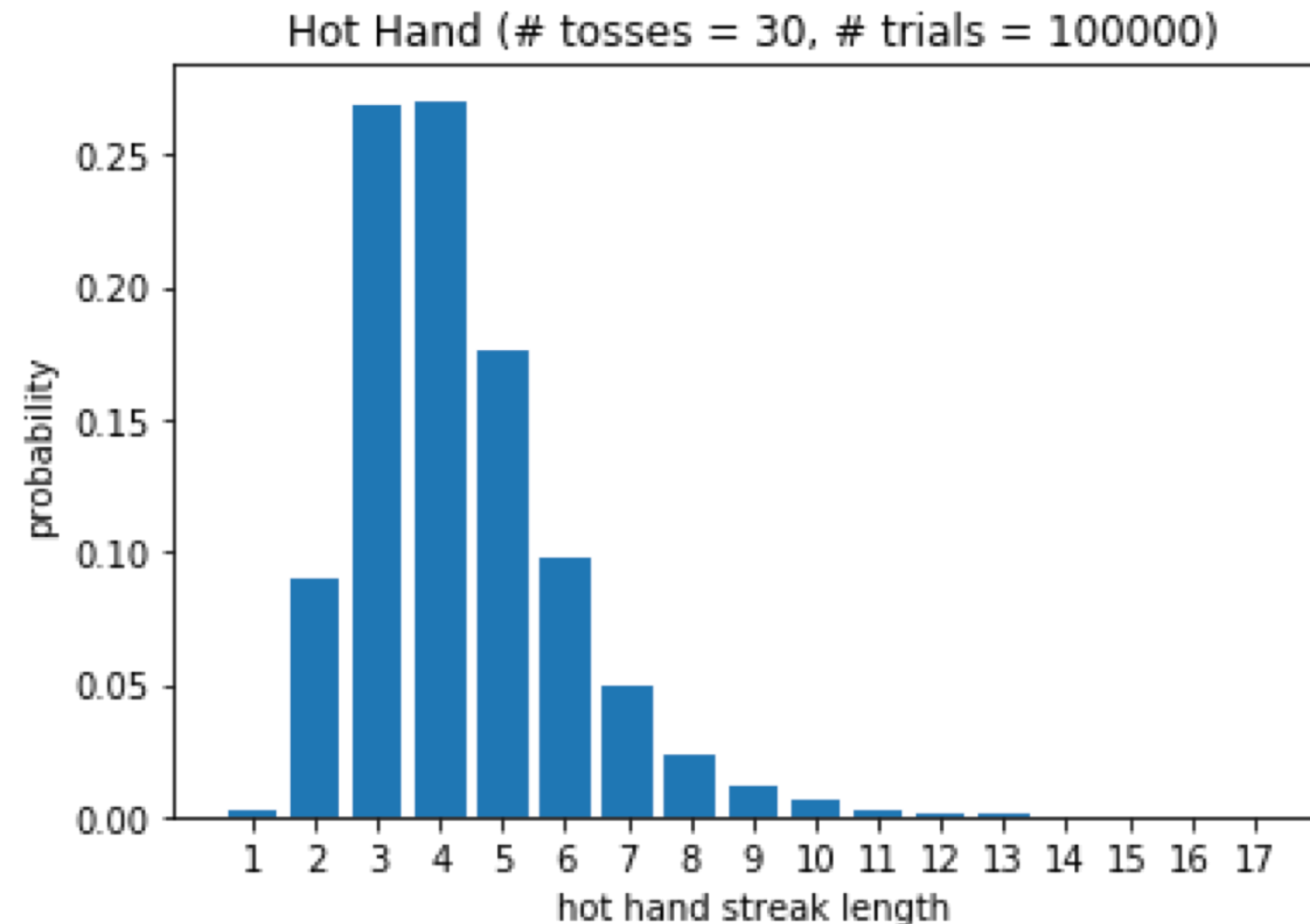
ASK THE AUDIENCE

- ▶ We toss a coin 50 times. Which event is most likely?

(A) HotHand = 2 (B) HotHand = 3

(C) HotHand = 4 (D) HotHand = 5

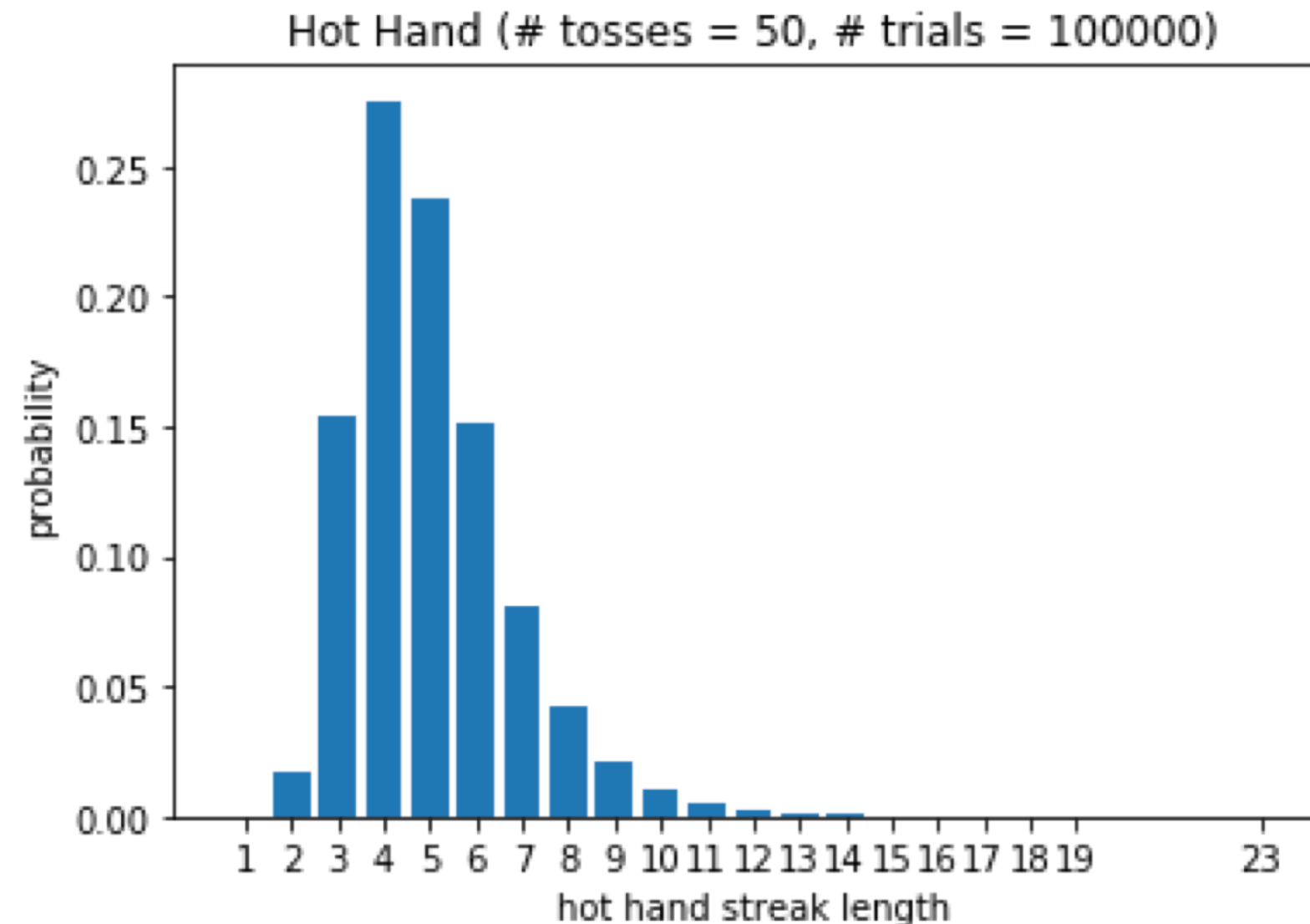
DISTRIBUTION FUNCTIONS



x-axis: possible values of the random variable

y-axis: Pr of taking on each value

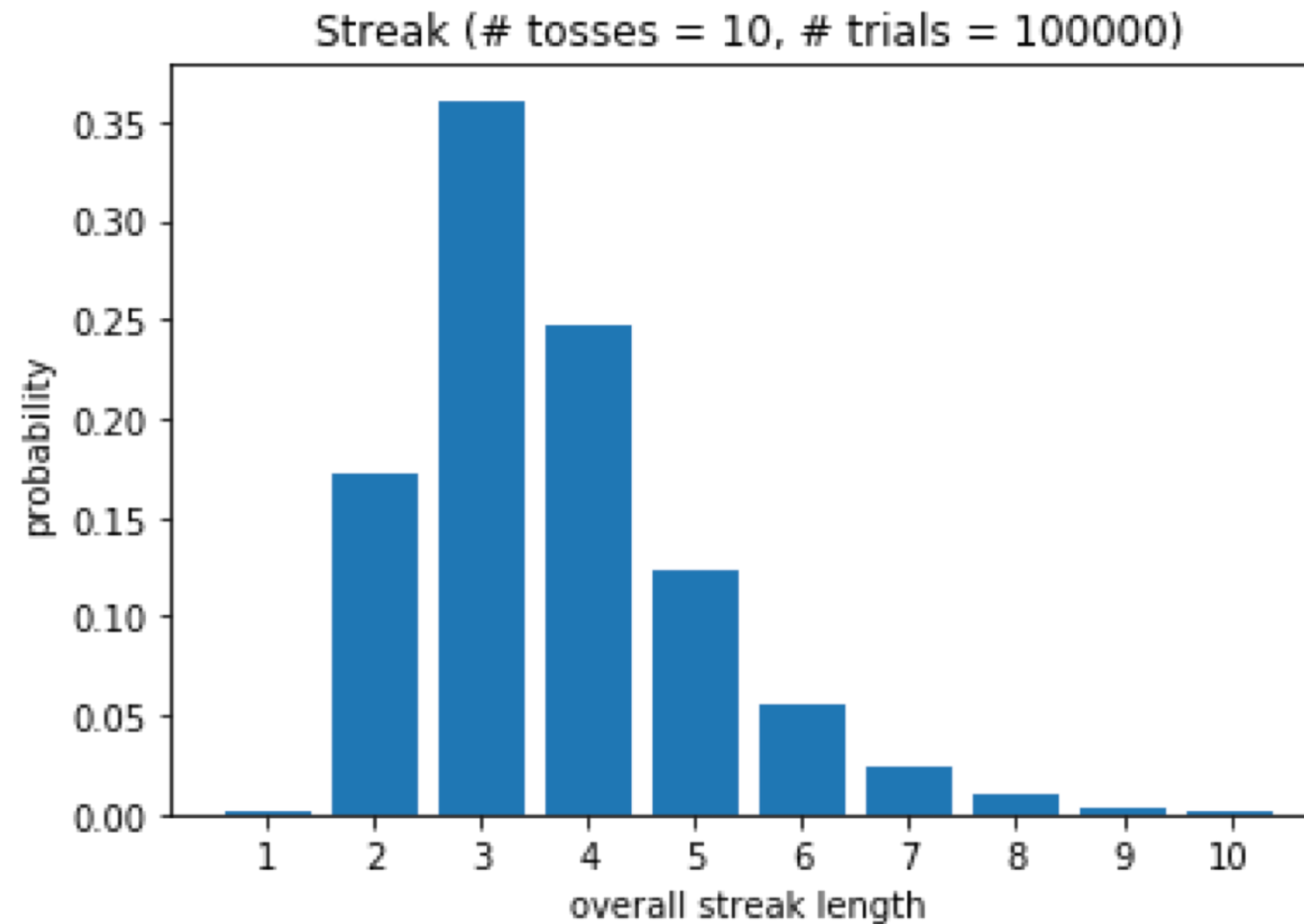
DISTRIBUTION FUNCTIONS



x-axis: possible values of the random variable

y-axis: Pr of taking on each value

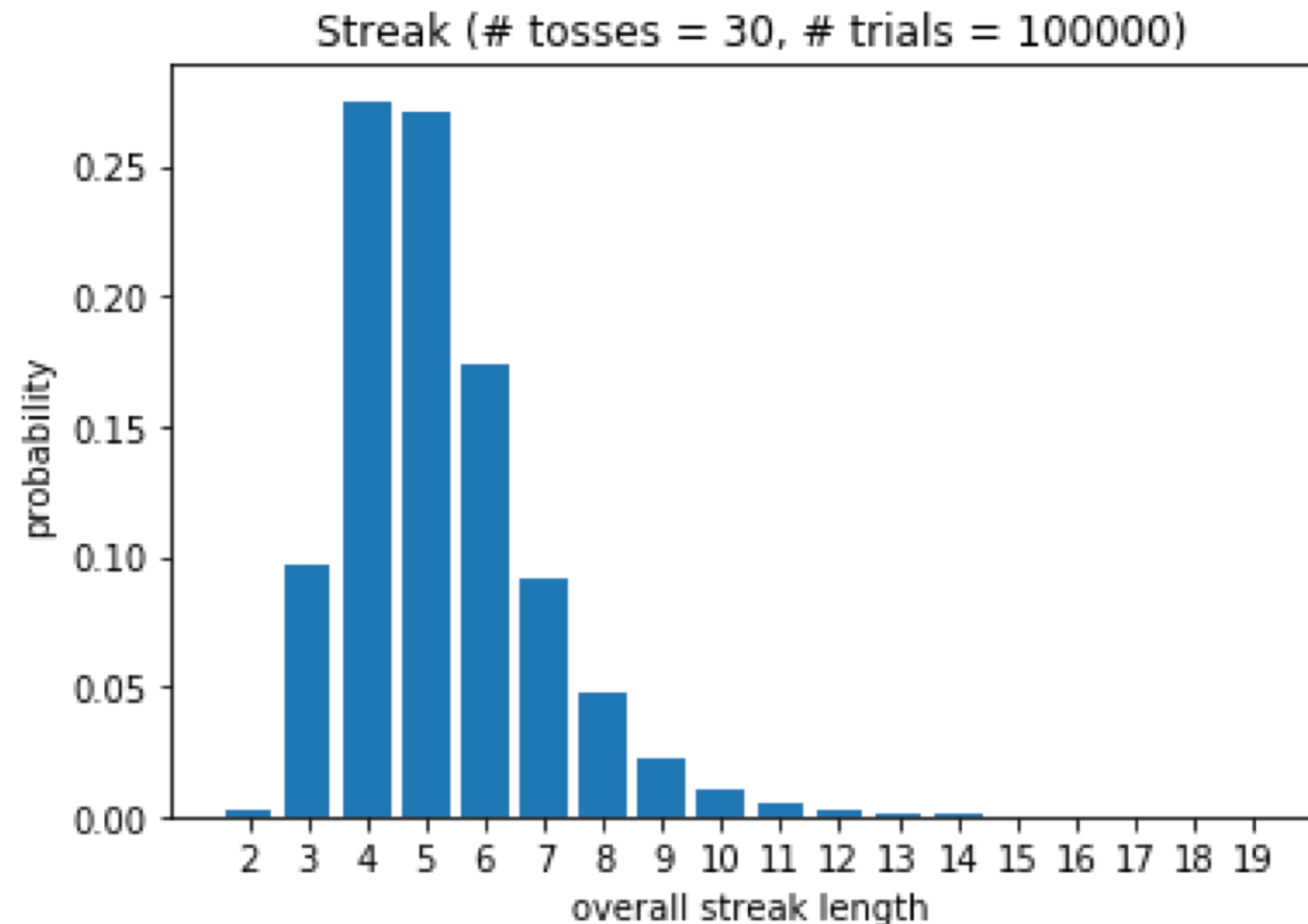
DISTRIBUTION FUNCTIONS



x-axis: possible values of the random variable

y-axis: Pr of taking on each value

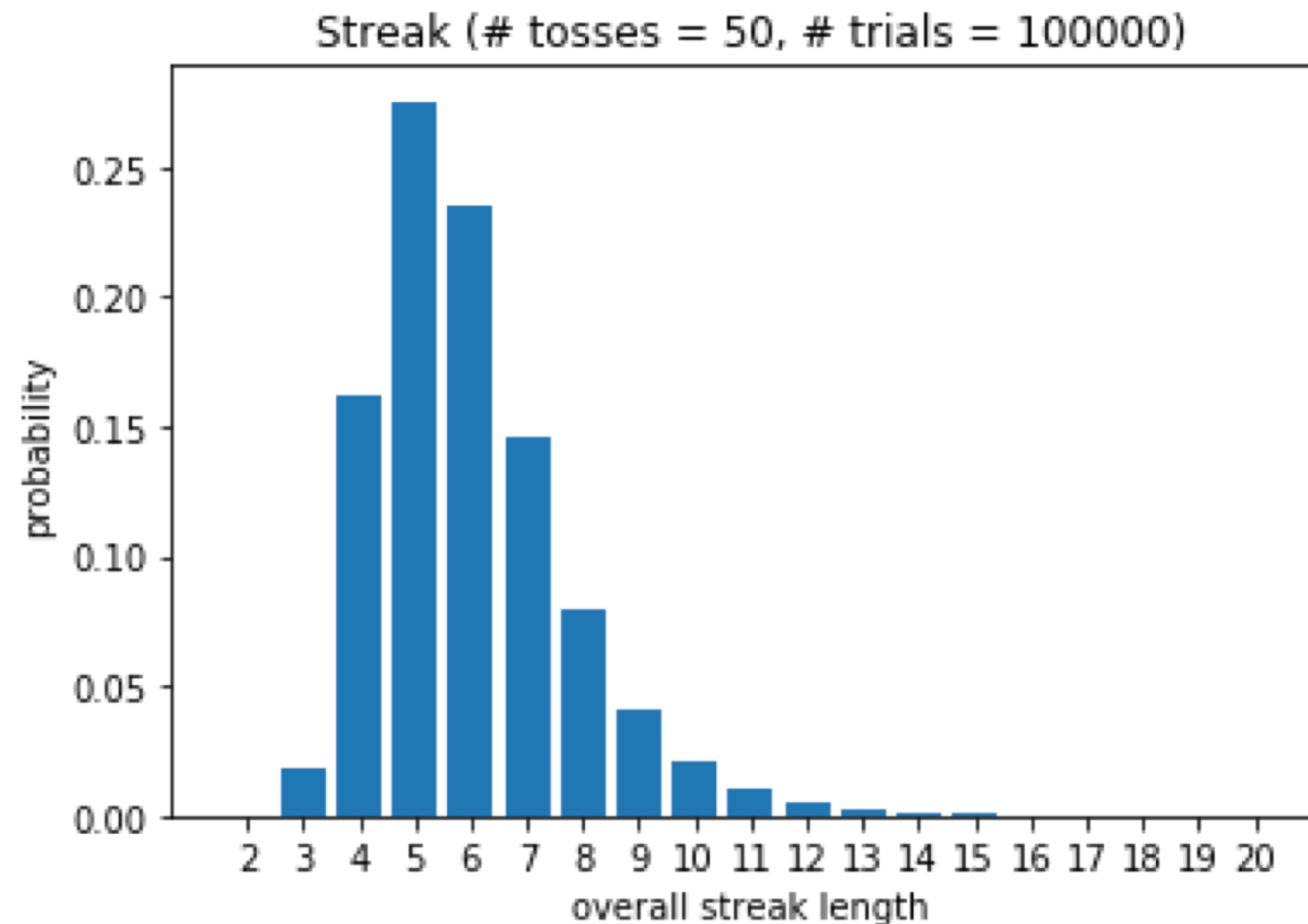
DISTRIBUTION FUNCTIONS



x-axis: possible values of the random variable

y-axis: Pr of taking on each value

DISTRIBUTION FUNCTIONS



x-axis: possible values of the random variable

y-axis: Pr of taking on each value

Sum of Random Variables

Functions of Random Variables

RANDOM VARIABLES: EXAMPLES

- **Experiment:** roll a six-sided die three times

$S = \text{the sum of the three rolls}$

$$S: \Omega \rightarrow \{3, 4, \dots, 18\}$$

$$S(1,1,1) = 3$$

$$S(1,1,2) = S(1,2,1) = S(2,1,1) = 4$$

...

$$S(6,6,6) = 18$$

$$Pr(S = 4) = ?$$

RANDOM VARIABLES: EXAMPLES

- **Experiment:** roll a six-sided die three times

$S = \text{the sum of the three rolls}$

$$S: \Omega \rightarrow \{3, 4, \dots, 18\}$$

$$S(1,1,1) = 3$$

$$S(1,1,2) = S(1,2,1) = S(2,1,1) = 4$$

...

$$S(6,6,6) = 18$$

$$Pr(S = 4) = \frac{|\{x \in \Omega \text{ such that } S(x) = 4\}|}{|\Omega|} = \frac{3}{216} = \frac{1}{72} \approx \frac{1}{6^3}$$

RANDOM VARIABLES

- **Experiment:** sample a value x uniformly from $[-1, 1]$

$X: [-1, 1] \rightarrow \mathbb{R}, X(r) = r$ continuous

$Y: [-1, 1] \rightarrow \mathbb{R}, Y(r) = r^2$ continuous

$Z: [-1, 1] \rightarrow \mathbb{R}, Z(r) = \begin{cases} 1 & \text{if } r > 0 \\ 0 & \text{if } r = 0 \\ -1 & \text{if } r < 0 \end{cases}$ discrete

DISTRIBUTION FUNCTIONS

- ▶ Let X be a **discrete** random variable
- ▶ The **probability mass function** (PMF) of X is the following function:

$$f_X: \mathbb{R} \rightarrow [0,1], f_X(k) = \Pr(X = k) \text{ for all } k \in \mathbb{R}$$

- ▶ The **cumulative density function** (CDF) of X is the following function:

$$F_X: \mathbb{R} \rightarrow [0,1], F_X(k) = \Pr(X \leq k) \text{ for all } k \in \mathbb{R}$$

$$F_X(k) = \Pr(X \leq k) = \sum_{k \leq x, \forall x \in \text{Range}(X)} \Pr(X=x)$$

outputs probabilities
for each value k in the
range of your random variable

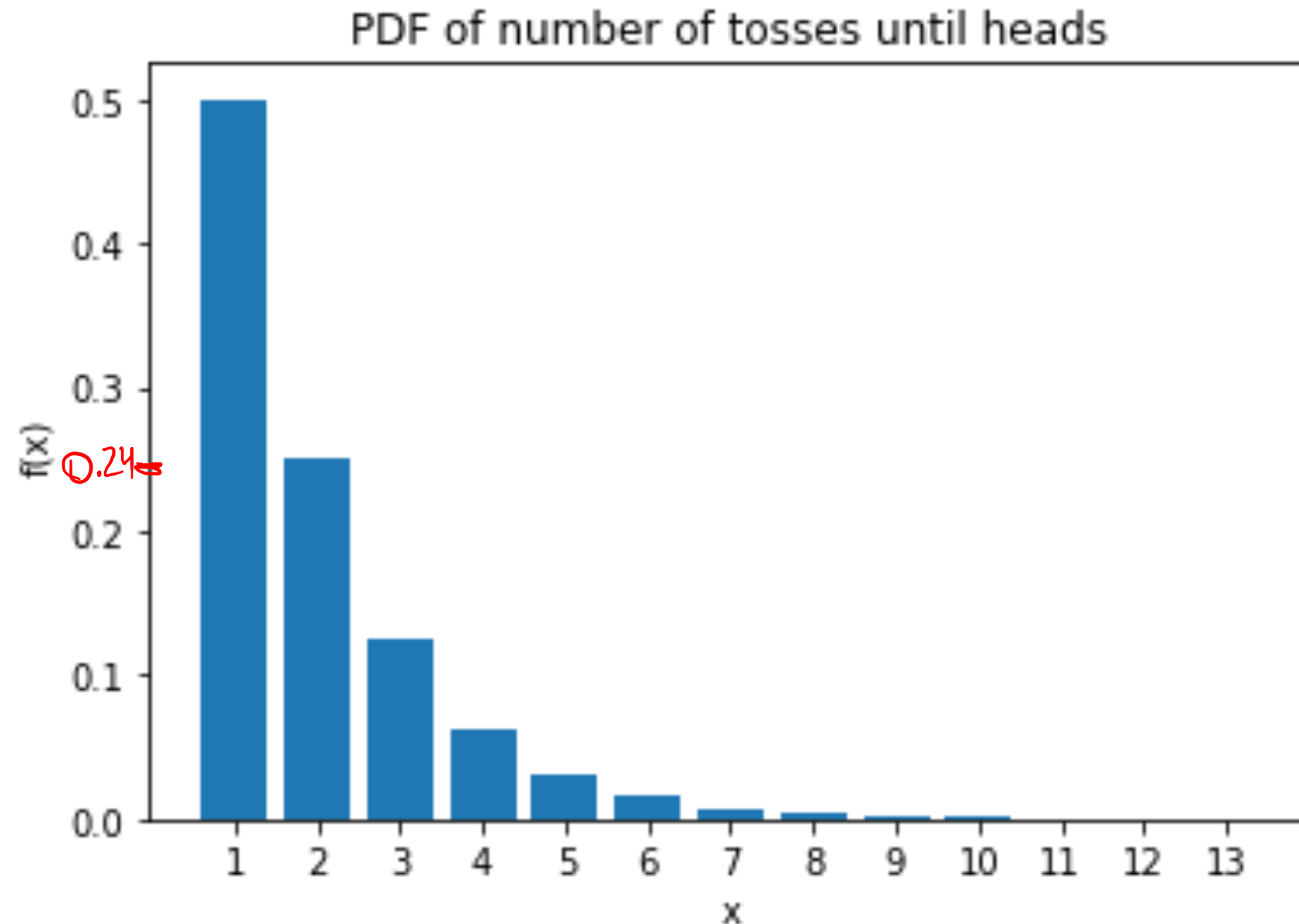
$k \in \text{Range}(X)$

sum of probabilities

DISTRIBUTION FUNCTIONS

$$\text{Range}(C) = \{1, 2, 3, 4, \dots\}$$

- ▶ Random variable: number of tosses until Heads



DISTRIBUTION FUNCTIONS

- Random variable: number of tosses until Heads

$$P_0(x \leq 1) = P_0(x = 1) = 0.5$$

$$P_0(x \leq 2) = 0.75$$

$$\begin{aligned} &= P_0(x = 1 \cup x = 2) \\ &= P_0(x = 1) + P_0(x = 2) \\ &= 0.5 + 0.25 \end{aligned}$$

